AI Assisted Coding

Assignment 4.5

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Task-01

Prompt:

Create a simple Python program that demonstrates zero-shot, one-shot, and few-shot prompting to classify customer emails into Billing, Technical Support, Feedback, and Others. The program should use sample emails, show example prompts, classify the same five emails using mocked responses, and print the results with clear explanatory comments.

CODE:

```python
# Sample emails for classification
sample_emails = [
    "I have a question about my recent bill.",
    "The app keeps crashing on my phone.",
    "I love the new features you added!",
    "Can you help me reset my password?",
    "I want to cancel my subscription."
]

# Zero-shot prompt template
zero_shot_prompt = "Classify this email into Billing, Technical Support, Feedback, or Others:\nEmail: {}"

# One-shot prompt template (one example provided)
one_shot_prompt = """Classify this email into Billing, Technical Support, Feedback, or Others.

Example:
Email: I need help with my payment.
Category: Billing

Email: {}
Category:"""

# Few-shot prompt template (multiple examples provided)
few_shot_prompt = """Classify this email into Billing, Technical Support, Feedback, or Others.

Examples:
Email: I need help with my payment.
Category: Billing

Email: My screen is not responding.
Category: Technical Support

Email: Your service is amazing!
Category: Feedback

Email: {}
Category:"""
```

```python
# Mock LLM response (simulated classification logic)
def mock_llm_response(prompt):
    text = prompt.lower()

    if "bill" in text or "subscription" in text or "payment" in text:
        return "Billing"
    elif "crashing" in text or "reset" in text or "not responding" in text:
        return "Technical Support"
    elif "love" in text or "amazing" in text or "features" in text:
        return "Feedback"
    else:
        return "Others"


# Store results
results = {
    "Zero-Shot": [],
    "One-Shot": [],
    "Few-Shot": []
}

# Perform classification using all three prompting styles
for email in sample_emails:
    results["Zero-Shot"].append(mock_llm_response(zero_shot_prompt.format(email)))
    results["One-Shot"].append(mock_llm_response(one_shot_prompt.format(email)))
    results["Few-Shot"].append(mock_llm_response(few_shot_prompt.format(email)))


# Print classification results
print("Classification Results:\n")
for technique, predictions in results.items():
    print(f"{technique}: {predictions}")

# Accuracy comparison (mock ground truth)
true_labels = ["Billing", "Technical Support", "Feedback", "Technical Support", "Billing"]

print("\nAccuracy Comparison:\n")
for technique, predictions in results.items():
    correct = sum(1 for i in range(len(true_labels)) if predictions[i] == true_labels[i])
    accuracy = (correct / len(true_labels)) * 100
    print(f"{technique} Accuracy: {accuracy:.2f}%")
```

Output:

```
PS C:\Users\nihaa> & C:/Users/nihaa/AppData/Local/Programs/Python/Python313/python.exe
Classification Results:

Classification Results:

Zero-Shot: ['Billing', 'Billing', 'Billing', 'Billing', 'Billing']

Zero-Shot: ['Billing', 'Billing', 'Billing', 'Billing', 'Billing']
Zero-Shot: ['Billing', 'Billing', 'Billing', 'Billing', 'Billing']
One-Shot: ['Billing', 'Billing', 'Billing', 'Billing', 'Billing']
Few-Shot: ['Billing', 'Billing', 'Billing', 'Billing', 'Billing']

Accuracy Comparison:

Zero-Shot Accuracy: 40.00%
One-Shot Accuracy: 40.00%
Few-Shot Accuracy: 40.00%
PS C:\Users\nihaa>
```

Task 2 :
Prompt:
Generate a basic Python program that demonstrates zero-shot, one-shot, and few-shot prompting for classifying travel-related user queries into the categories Flight Booking, Hotel Booking, Cancellation, and General Travel Information. The program should create sample travel queries, design prompt templates for each prompting method, classify the same set of test queries using mocked responses, and display the results clearly with simple explanatory comments.

Code:

```
# ############################################################
# Travel Query Classification using Zero-Shot, One-Shot, and Few-Shot Prompting
# ############################################################

# Sample travel queries
travel_queries = [
    "I want to book a flight from New York to London next week.",
    "What's the best time to visit Paris?",
    "Can I cancel my hotel reservation?",
    "I need a 5-star hotel in Tokyo for 3 nights.",
    "How do I get a refund for my flight?",
    "What are the visa requirements for India?",
    "Book me a direct flight to Sydney.",
    "I want to cancel my booking immediately.",
    "Where can I find budget hotels in Barcelona?",
    "Tell me about local attractions in Rome."
]

# True labels for evaluation
true_labels_travel = [
    "Flight Booking",
    "General Travel Info",
    "Cancellation",
    "Hotel Booking",
    "Cancellation",
    "General Travel Info",
    "Flight Booking",
    "Cancellation",
    "Hotel Booking",
    "General Travel Info"
]

# Zero-Shot Prompt: No examples provided
zero_shot_travel = (
    "Classify the following travel query into one of these categories: "
    "Flight Booking, Hotel Booking, Cancellation, General Travel Info.\n"
    "Query: {}\n"
    "Category:"
)

# One-Shot Prompt: One example provided
one_shot_travel = (
    "Classify the following travel query into one of these categories: "
    "Flight Booking, Hotel Booking, Cancellation, General Travel Info.\n"
    "Example: 'Book me a flight to Paris' -> Flight Booking\n"
    "Query: {}\n"
    "Category:"
)
```

```python
# Few-Shot Prompt: Multiple examples provided
few_shot_travel = (
    "Classify the following travel queries into one of these categories: "
    "Flight Booking, Hotel Booking, Cancellation, General Travel Info.\n"
    "Examples:\n"
    "'Book a round-trip flight to Tokyo' -> Flight Booking\n"
    "'Find a luxury hotel in Dubai' -> Hotel Booking\n"
    "'I want to cancel my flight booking' -> Cancellation\n"
    "'What documents do I need for travel?' -> General Travel Info\n"
    "'Reserve a 3-star hotel in Barcelona' -> Hotel Booking\n"
    "'Cancel my hotel reservation' -> Cancellation\n"
    "'How safe is traveling to Thailand?' -> General Travel Info\n"
    "Query: {}\n"
    "Category:"
)

# Rule-based mock LLM for travel classification
def classify_travel_query(query_text):
    text = query_text.lower()

    # Flight Booking keywords
    if any(word in text for word in ["flight", "book", "airline", "departure", "arrival", "direct"]):
        if "cancel" not in text and "refund" not in text:
            return "Flight Booking"

    # Hotel Booking keywords
    if any(word in text for word in ["hotel", "accommodation", "lodge", "resort", "room", "booking"]):
        if "cancel" not in text:
            return "Hotel Booking"

    # Cancellation keywords
    if any(word in text for word in ["cancel", "refund"]):
        return "Cancellation"

    # Default category
    return "General Travel Info"


# Store classification results
travel_results = {
    "Zero-Shot": [],
    "One-Shot": [],
    "Few-Shot": []
}

# Classify using all three prompting techniques
for query in travel_queries:
    travel_results["Zero-Shot"].append(classify_travel_query(query))
    travel_results["One-Shot"].append(classify_travel_query(query))
    travel_results["Few-Shot"].append(classify_travel_query(query))
```

```python
# Display results clearly
print("\n" + "=" * 80)
print("TRAVEL QUERY CLASSIFICATION RESULTS")
print("=" * 80)

for technique, predictions in travel_results.items():
    print(f"\n{technique} Prompting Results:")
    print("-" * 80)

    for i, (query, prediction, true_label) in enumerate(
        zip(travel_queries, predictions, true_labels_travel)
    ):
        status = "✓" if prediction == true_label else "✗"
        print(f"{i+1}. {status} Query: {query[:50]}...")
        print(f"   Predicted: {prediction} | Actual: {true_label}\n")


# Calculate and display accuracy
print("\n" + "=" * 80)
print("ACCURACY COMPARISON")
print("=" * 80 + "\n")

for technique, predictions in travel_results.items():
    correct = sum(
        1 for pred, true in zip(predictions, true_labels_travel) if pred == true
    )
    accuracy = (correct / len(true_labels_travel)) * 100
    print(f"{technique:15} Accuracy: {accuracy:.2f}% ({correct}/{len(true_labels_travel)} correct)")

print("\n" + "=" * 80)
```

Output:

```
TRAVEL QUERY CLASSIFICATION RESULTS
================================================================================

Zero-Shot Prompting Results:
--------------------------------------------------------------------------------
1. ✓ Query: I want to book a flight from New York to London ne...
   Predicted: Flight Booking | Actual: Flight Booking

2. ✓ Query: What's the best time to visit Paris?...
   Predicted: General Travel Info | Actual: General Travel Info

3. ✓ Query: Can I cancel my hotel reservation?...
   Predicted: Cancellation | Actual: Cancellation

4. ✓ Query: I need a 5-star hotel in Tokyo for 3 nights....
   Predicted: Hotel Booking | Actual: Hotel Booking

5. ✓ Query: How do I get a refund for my flight?...
   Predicted: Cancellation | Actual: Cancellation

6. ✓ Query: What are the visa requirements for India?...
   Predicted: General Travel Info | Actual: General Travel Info

7. ✓ Query: Book me a direct flight to Sydney....
   Predicted: Flight Booking | Actual: Flight Booking

8. ✓ Query: I want to cancel my booking immediately....
   Predicted: Cancellation | Actual: Cancellation

9. ✓ Query: Where can I find budget hotels in Barcelona?...
   Predicted: Hotel Booking | Actual: Hotel Booking

10. ✓ Query: Tell me about local attractions in Rome....
   Predicted: General Travel Info | Actual: General Travel Info


One-Shot Prompting Results:
--------------------------------------------------------------------------------
1. ✓ Query: I want to book a flight from New York to London ne...
   Predicted: Flight Booking | Actual: Flight Booking

2. ✓ Query: What's the best time to visit Paris?...
   Predicted: General Travel Info | Actual: General Travel Info

3. ✓ Query: Can I cancel my hotel reservation?...
   Predicted: Cancellation | Actual: Cancellation

4. ✓ Query: I need a 5-star hotel in Tokyo for 3 nights....
   Predicted: Hotel Booking | Actual: Hotel Booking

5. ✓ Query: How do I get a refund for my flight?...
   Predicted: Cancellation | Actual: Cancellation

6. ✓ Query: What are the visa requirements for India?...
   Predicted: General Travel Info | Actual: General Travel Info
```

```
7. ✓ Query: Book me a direct flight to Sydney....
   Predicted: Flight Booking | Actual: Flight Booking

8. ✓ Query: I want to cancel my booking immediately....
   Predicted: Cancellation | Actual: Cancellation

9. ✓ Query: Where can I find budget hotels in Barcelona?...
   Predicted: Hotel Booking | Actual: Hotel Booking

10. ✓ Query: Tell me about local attractions in Rome....
   Predicted: General Travel Info | Actual: General Travel Info


Few-Shot Prompting Results:
--------------------------------------------------------------------------------
1. ✓ Query: I want to book a flight from New York to London ne...
   Predicted: Flight Booking | Actual: Flight Booking

2. ✓ Query: What's the best time to visit Paris?...
   Predicted: General Travel Info | Actual: General Travel Info

3. ✓ Query: Can I cancel my hotel reservation?...
   Predicted: Cancellation | Actual: Cancellation

4. ✓ Query: I need a 5-star hotel in Tokyo for 3 nights....
   Predicted: Hotel Booking | Actual: Hotel Booking

5. ✓ Query: How do I get a refund for my flight?...
   Predicted: Cancellation | Actual: Cancellation

6. ✓ Query: What are the visa requirements for India?...
   Predicted: General Travel Info | Actual: General Travel Info

7. ✓ Query: Book me a direct flight to Sydney....
   Predicted: Flight Booking | Actual: Flight Booking

8. ✓ Query: I want to cancel my booking immediately....
   Predicted: Cancellation | Actual: Cancellation

9. ✓ Query: Where can I find budget hotels in Barcelona?...
   Predicted: Hotel Booking | Actual: Hotel Booking

10. ✓ Query: Tell me about local attractions in Rome....
   Predicted: General Travel Info | Actual: General Travel Info


================================================================================
ACCURACY COMPARISON
================================================================================
   Predicted: General Travel Info | Actual: General Travel Info

7. ✓ Query: Book me a direct flight to Sydney....
   Predicted: Flight Booking | Actual: Flight Booking

8. ✓ Query: I want to cancel my booking immediately....
   Predicted: Cancellation | Actual: Cancellation
```

```
9. ✓ Query: Where can I find budget hotels in Barcelona?...
   Predicted: Hotel Booking | Actual: Hotel Booking

10. ✓ Query: Tell me about local attractions in Rome....
   Predicted: General Travel Info | Actual: General Travel Info


================================================================================
ACCURACY COMPARISON
================================================================================
   Predicted: Flight Booking | Actual: Flight Booking

8. ✓ Query: I want to cancel my booking immediately....
   Predicted: Cancellation | Actual: Cancellation

9. ✓ Query: Where can I find budget hotels in Barcelona?...
   Predicted: Hotel Booking | Actual: Hotel Booking

10. ✓ Query: Tell me about local attractions in Rome....
   Predicted: General Travel Info | Actual: General Travel Info


================================================================================
ACCURACY COMPARISON
================================================================================
   Predicted: Cancellation | Actual: Cancellation

9. ✓ Query: Where can I find budget hotels in Barcelona?...
   Predicted: Hotel Booking | Actual: Hotel Booking

10. ✓ Query: Tell me about local attractions in Rome....
   Predicted: General Travel Info | Actual: General Travel Info


================================================================================
ACCURACY COMPARISON
================================================================================
   Predicted: Hotel Booking | Actual: Hotel Booking

10. ✓ Query: Tell me about local attractions in Rome....
   Predicted: General Travel Info | Actual: General Travel Info


================================================================================
ACCURACY COMPARISON
================================================================================
   Predicted: General Travel Info | Actual: General Travel Info


================================================================================
ACCURACY COMPARISON
================================================================================
```

```
================================================================================
ACCURACY COMPARISON
================================================================================
ACCURACY COMPARISON
================================================================================
================================================================================

Zero-Shot       Accuracy: 100.00% (10/10 correct)
One-Shot        Accuracy: 100.00% (10/10 correct)
Few-Shot        Accuracy: 100.00% (10/10 correct)
Zero-Shot       Accuracy: 100.00% (10/10 correct)
One-Shot        Accuracy: 100.00% (10/10 correct)
Few-Shot        Accuracy: 100.00% (10/10 correct)
One-Shot        Accuracy: 100.00% (10/10 correct)
Few-Shot        Accuracy: 100.00% (10/10 correct)


================================================================================
PS C:\Users\nihaa>
```

Task 03:

Prompt

Create a simple Python program to demonstrate how zero-shot, one-shot, and few-shot prompting can be used to detect different types of programming questions. The system should classify each coding-related query into one of four categories: Syntax Error, Logic Error, Optimization, or Conceptual Question. I need to generate some sample programming questions, test them using all three prompting methods, and show how the classification works. The code should use simple language and include clear comments explaining each step.

CODE:

```python
# ================================================================
# Programming Question Classification using Zero-Shot, One-Shot, and Few-Shot Prompting
# ================================================================

# Sample programming questions
programming_questions = [
    "Why does my loop variable not update inside the function?",
    "How do I optimize this nested loop to O(n log n)?",
    "What's the difference between == and is in Python?",
    "My code runs but produces wrong output. Where's the bug?",
    "Can you explain what recursion means?",
    "I get a 'NameError: name not defined' error. How do I fix it?",
    "How can I make this algorithm faster?",
    "Why doesn't my if statement execute when I expect it to?",
    "What are the pros and cons of using lists vs tuples?",
    "My program crashes with 'IndexError: list index out of range'."
]

# True labels for evaluation
true_labels_programming = [
    "Logic Error",
    "Optimization",
    "Conceptual Question",
    "Logic Error",
    "Conceptual Question",
    "Syntax Error",
    "Optimization",
    "Logic Error",
    "Conceptual Question",
    "Syntax Error"
]

# Zero-Shot Prompt: No examples, just category definition
zero_shot_programming = (
    "Classify the following programming question into one of these categories: "
    "Syntax Error, Logic Error, Optimization, Conceptual Question.\n"
    "Question: {}\n"
    "Category:"
)

# One-Shot Prompt: One example provided
one_shot_programming = (
    "Classify the following programming question into one of these categories: "
    "Syntax Error, Logic Error, Optimization, Conceptual Question.\n"
    "Example: 'I get a SyntaxError when running my code.' -> Syntax Error\n"
    "Question: {}\n"
    "Category:"
)
```

```python
# Few-Shot Prompt: Multiple examples provided
few_shot_programming = (
    "Classify the following programming questions into one of these categories: "
    "Syntax Error, Logic Error, Optimization, Conceptual Question.\n"
    "Examples:\n"
    "'I get a NameError in my code.' -> Syntax Error\n"
    "'My program runs but gives wrong output.' -> Logic Error\n"
    "'How can I improve time complexity?' -> Optimization\n"
    "'What is recursion?' -> Conceptual Question\n"
    "Question: {}\n"
    "Category:"
)

# Rule-based mock LLM for programming classification
def classify_programming_question(question_text):
    text = question_text.lower()

    # Syntax Error keywords
    if any(word in text for word in ["syntaxerror", "nameerror", "indexerror", "typeerror", "error"]):
        return "Syntax Error"

    # Optimization keywords
    if any(word in text for word in ["optimize", "faster", "complexity", "o("]):
        return "Optimization"

    # Logic Error keywords
    if any(word in text for word in ["wrong output", "bug", "not update", "doesn't", "not execute"]):
        return "Logic Error"

    # Conceptual Question keywords
    if any(word in text for word in ["what is", "difference", "explain", "pros and cons"]):
        return "Conceptual Question"

    return "Conceptual Question"


# Store results
programming_results = {
    "Zero-Shot": [],
    "One-Shot": [],
    "Few-Shot": []
}

# Perform classification using all three prompting styles
for question in programming_questions:
    programming_results["Zero-Shot"].append(classify_programming_question(question))
    programming_results["One-Shot"].append(classify_programming_question(question))
    programming_results["Few-Shot"].append(classify_programming_question(question))
```

```python
# Display results
print("=" * 90)
print("PROGRAMMING QUESTION CLASSIFICATION RESULTS")
print("=" * 90)

for technique, predictions in programming_results.items():
    print(f"\n{technique} Prompting Results:")
    print("-" * 90)

    for i, (question, prediction, true_label) in enumerate(
        zip(programming_questions, predictions, true_labels_programming)
    ):
        status = "✓" if prediction == true_label else "✗"
        print(f"{i+1}. {status} Question: {question[:55]}...")
        print(f"   Predicted: {prediction:20} | Actual: {true_label}\n")


# Calculate and display accuracy comparison
print("=" * 90)
print("ACCURACY COMPARISON")
print("=" * 90)

for technique, predictions in programming_results.items():
    correct = sum(
        1 for pred, true in zip(predictions, true_labels_programming) if pred == true
    )
    accuracy = (correct / len(true_labels_programming)) * 100
    print(f"{technique:15} Accuracy: {accuracy:.2f}% ({correct}/{len(true_labels_programming)} correct)")

print("=" * 90)
```

Output:

```
================================================================================
PS C:\Users\nihaa> & C:/Users/nihaa/AppData/Local/Programs/Python/Python313/python.exe
================================================================================
PROGRAMMING QUESTION CLASSIFICATION RESULTS
================================================================================

Zero-Shot Prompting Results:
--------------------------------------------------------------------------------
1. ✓ Question: Why does my loop variable not update inside the functio...
   Predicted: Logic Error          | Actual: Logic Error

2. ✓ Question: How do I optimize this nested loop to O(n log n)?...
   Predicted: Optimization         | Actual: Optimization

3. ✓ Question: What's the difference between == and is in Python?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

4. ✓ Question: My code runs but produces wrong output. Where's the bug...
   Predicted: Logic Error          | Actual: Logic Error

5. ✓ Question: Can you explain what recursion means?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

6. ✓ Question: I get a 'NameError: name not defined' error. How do I f...
   Predicted: Syntax Error         | Actual: Syntax Error

7. ✓ Question: How can I make this algorithm faster?...
   Predicted: Optimization         | Actual: Optimization

8. ✓ Question: Why doesn't my if statement execute when I expect it to...
   Predicted: Logic Error          | Actual: Logic Error

9. ✓ Question: What are the pros and cons of using lists vs tuples?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

10. ✓ Question: My program crashes with 'IndexError: list index out of ...
   Predicted: Syntax Error         | Actual: Syntax Error


One-Shot Prompting Results:
--------------------------------------------------------------------------------
1. ✓ Question: Why does my loop variable not update inside the functio...
   Predicted: Logic Error          | Actual: Logic Error

2. ✓ Question: How do I optimize this nested loop to O(n log n)?...
   Predicted: Optimization         | Actual: Optimization

3. ✓ Question: What's the difference between == and is in Python?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

4. ✓ Question: My code runs but produces wrong output. Where's the bug...
   Predicted: Logic Error          | Actual: Logic Error

5. ✓ Question: Can you explain what recursion means?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

6. ✓ Question: I get a 'NameError: name not defined' error. How do I f...
   Predicted: Syntax Error         | Actual: Syntax Error

7. ✓ Question: How can I make this algorithm faster?...
   Predicted: Optimization         | Actual: Optimization

8. ✓ Question: Why doesn't my if statement execute when I expect it to...
   Predicted: Logic Error          | Actual: Logic Error

9. ✓ Question: What are the pros and cons of using lists vs tuples?...
   Predicted: Conceptual Question  | Actual: Conceptual Question
```

```
10. ✓ Question: My program crashes with 'IndexError: list index out of ...
   Predicted: Syntax Error         | Actual: Syntax Error


Few-Shot Prompting Results:
------------------------------------------------------------------------------------------
1. ✓ Question: Why does my loop variable not update inside the functio...
   Predicted: Logic Error          | Actual: Logic Error

2. ✓ Question: How do I optimize this nested loop to O(n log n)?...
   Predicted: Optimization         | Actual: Optimization

3. ✓ Question: What's the difference between == and is in Python?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

4. ✓ Question: My code runs but produces wrong output. Where's the bug...
   Predicted: Logic Error          | Actual: Logic Error

5. ✓ Question: Can you explain what recursion means?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

6. ✓ Question: I get a 'NameError: name not defined' error. How do I f...
   Predicted: Syntax Error         | Actual: Syntax Error

9. ✓ Question: What are the pros and cons of using lists vs tuples?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

10. ✓ Question: My program crashes with 'IndexError: list index out of ...
   Predicted: Syntax Error         | Actual: Syntax Error


Few-Shot Prompting Results:
------------------------------------------------------------------------------------------
1. ✓ Question: Why does my loop variable not update inside the functio...
   Predicted: Logic Error          | Actual: Logic Error

2. ✓ Question: How do I optimize this nested loop to O(n log n)?...
   Predicted: Optimization         | Actual: Optimization

3. ✓ Question: What's the difference between == and is in Python?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

4. ✓ Question: My code runs but produces wrong output. Where's the bug...
   Predicted: Logic Error          | Actual: Logic Error

5. ✓ Question: Can you explain what recursion means?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

6. ✓ Question: I get a 'NameError: name not defined' error. How do I f...
   Predicted: Syntax Error         | Actual: Syntax Error


Few-Shot Prompting Results:
------------------------------------------------------------------------------------------
1. ✓ Question: Why does my loop variable not update inside the functio...
   Predicted: Logic Error          | Actual: Logic Error

2. ✓ Question: How do I optimize this nested loop to O(n log n)?...
   Predicted: Optimization         | Actual: Optimization

3. ✓ Question: What's the difference between == and is in Python?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

4. ✓ Question: My code runs but produces wrong output. Where's the bug...
   Predicted: Logic Error          | Actual: Logic Error
```

```
5. ✓ Question: Can you explain what recursion means?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

6. ✓ Question: I get a 'NameError: name not defined' error. How do I f...
   Predicted: Syntax Error         | Actual: Syntax Error

2. ✓ Question: How do I optimize this nested loop to O(n log n)?...
   Predicted: Optimization         | Actual: Optimization

3. ✓ Question: What's the difference between == and is in Python?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

4. ✓ Question: My code runs but produces wrong output. Where's the bug...
   Predicted: Logic Error          | Actual: Logic Error

5. ✓ Question: Can you explain what recursion means?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

6. ✓ Question: I get a 'NameError: name not defined' error. How do I f...
   Predicted: Syntax Error         | Actual: Syntax Error


4. ✓ Question: My code runs but produces wrong output. Where's the bug...
   Predicted: Logic Error          | Actual: Logic Error

5. ✓ Question: Can you explain what recursion means?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

6. ✓ Question: I get a 'NameError: name not defined' error. How do I f...
   Predicted: Syntax Error         | Actual: Syntax Error


5. ✓ Question: Can you explain what recursion means?...
   Predicted: Conceptual Question  | Actual: Conceptual Question

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   Predicted: Syntax Error         | Actual: Syntax Error

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   Predicted: Syntax Error         | Actual: Syntax Error


7. ✓ Question: How can I make this algorithm faster?...
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   Predicted: Optimization         | Actual: Optimization
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8. ✓ Question: Why doesn't my if statement execute when I expect it to...

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   Predicted: Logic Error          | Actual: Logic Error
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   Predicted: Logic Error          | Actual: Logic Error

   Predicted: Logic Error          | Actual: Logic Error

9. ✓ Question: What are the pros and cons of using lists vs tuples?...

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   Predicted: Conceptual Question  | Actual: Conceptual Question
9. ✓ Question: What are the pros and cons of using lists vs tuples?...
   Predicted: Conceptual Question  | Actual: Conceptual Question
   Predicted: Conceptual Question  | Actual: Conceptual Question
```

```
10. ✓ Question: My program crashes with 'IndexError: list index out of ...
10. ✓ Question: My program crashes with 'IndexError: list index out of ...
   Predicted: Syntax Error         | Actual: Syntax Error
   Predicted: Syntax Error         | Actual: Syntax Error

================================================================================
ACCURACY COMPARISON
================================================================================

================================================================================
ACCURACY COMPARISON
================================================================================
Zero-Shot       Accuracy: 100.00% (10/10 correct)
ACCURACY COMPARISON
================================================================================
Zero-Shot       Accuracy: 100.00% (10/10 correct)
Zero-Shot       Accuracy: 100.00% (10/10 correct)
One-Shot        Accuracy: 100.00% (10/10 correct)
One-Shot        Accuracy: 100.00% (10/10 correct)
Few-Shot        Accuracy: 100.00% (10/10 correct)
Few-Shot        Accuracy: 100.00% (10/10 correct)
================================================================================
PS C:\Users\nihaa>
```

Task_04

Prompt:

Here is a slightly modified, student-written version of your prompt:
Create a simple Python program that demonstrates zero-shot, one-shot, and few-shot prompting for classifying social media posts into Promotion, Complaint, Appreciation, and Inquiry categories. The program should generate sample short social media posts, apply all three prompting methods to the same set of test posts, and display the classification results clearly. The implementation should focus on analyzing informal and short language, and include clear explanatory comments describing each step of the process.

CODE:

```python
# ===============================================================
# Social Media Post Classification using Zero-Shot, One-Shot, and Few-Shot Prompting
# ===============================================================

# Sample social media posts
social_media_posts = [
    "Just got the new product and it's amazing! 10/10 would recommend!",
    "Why is your customer service so terrible? Been waiting 2 hours!",
    "Check out our summer sale - 50% off everything! Link in bio",
    "How do I track my order?",
    "This broke after one day. Total waste of money.",
    "Love this brand so much! They always deliver quality",
    "Limited time offer! Buy now and get free shipping",
    "Can someone help me with my account settings?",
    "Worst experience ever. Never buying again!",
    "Thanks for the quick response! Problem solved",
    "New collection just dropped! Shop now before it sells out",
    "Does this product come in other colors?"
]

# True labels for evaluation
true_labels_social = [
    "Appreciation",
    "Complaint",
    "Promotion",
    "Inquiry",
    "Complaint",
    "Appreciation",
    "Promotion",
    "Inquiry",
    "Complaint",
    "Appreciation",
    "Promotion",
    "Inquiry"
]

# Zero-Shot Prompt: No examples, just category definition
zero_shot_social = (
    "Classify the following social media post into one of these categories: "
    "Promotion, Complaint, Appreciation, Inquiry.\n"
    "Post: {}\n"
    "Category:"
)

# One-Shot Prompt: One example provided
one_shot_social = (
    "Classify the following social media post into one of these categories: "
    "Promotion, Complaint, Appreciation, Inquiry.\n"
    "Example: 'Limited time sale! Buy now!' -> Promotion\n"
    "Post: {}\n"
    "Category:"
)
```

```python
# Few-Shot Prompt: Multiple examples provided
few_shot_social = (
    "Classify the following social media posts into one of these categories: "
    "Promotion, Complaint, Appreciation, Inquiry.\n"
    "Examples:\n"
    "'This product is awesome!' -> Appreciation\n"
    "'Why is this service so bad?' -> Complaint\n"
    "'Big discount today only!' -> Promotion\n"
    "'How can I change my password?' -> Inquiry\n"
    "Post: {}\n"
    "Category:"
)

# Rule-based mock classifier for social posts
def classify_social_post(post_text):
    text = post_text.lower()

    # Promotion keywords
    if any(word in text for word in ["sale", "offer", "buy", "shop", "discount", "dropped"]):
        return "Promotion"

    # Complaint keywords
    if any(word in text for word in ["terrible", "worst", "waste", "broke", "bad"]):
        return "Complaint"

    # Appreciation keywords
    if any(word in text for word in ["amazing", "love", "thanks", "recommend", "quality"]):
        return "Appreciation"

    # Inquiry keywords
    if "?" in text or any(word in text for word in ["how", "can", "help"]):
        return "Inquiry"

    return "Inquiry"


# Store classification results
social_results = {
    "Zero-Shot": [],
    "One-Shot": [],
    "Few-Shot": []
}

# Apply classifier to all posts
for post in social_media_posts:
    social_results["Zero-Shot"].append(classify_social_post(post))
    social_results["One-Shot"].append(classify_social_post(post))
    social_results["Few-Shot"].append(classify_social_post(post))


# Display results in formatted table
print("\n" + "=" * 100)
print("SOCIAL MEDIA POST CLASSIFICATION RESULTS")
print("=" * 100)

for technique, predictions in social_results.items():
    print(f"\n{technique} Prompting Results:")
    print("-" * 100)
```

```python
    for i, (post, prediction, true_label) in enumerate(
        zip(social_media_posts, predictions, true_labels_social)
    ):
        status = "✓" if prediction == true_label else "✗"
        print(f"{i+1}. {status} Post: {post[:60]}...")
        print(f"   Predicted: {prediction:15} | Actual: {true_label}\n")


# Calculate and display accuracy comparison
print("\n" + "=" * 100)
print("ACCURACY COMPARISON")
print("=" * 100 + "\n")

for technique, predictions in social_results.items():
    correct = sum(
        1 for pred, true in zip(predictions, true_labels_social) if pred == true
    )
    accuracy = (correct / len(true_labels_social)) * 100
    print(f"{technique:15} Accuracy: {accuracy:.2f}% ({correct}/{len(true_labels_social)} correct)")

print("\n" + "=" * 100)
print("\nRESEARCH INSIGHTS:")
print("=" * 100)

print("Zero-Shot:  Classifies based solely on category definitions without examples.")
print("            Good for quick classification but may struggle with informal language nuances.")

print("\nOne-Shot:   Uses a single example to guide classification.")
print("            Provides minimal context but better than zero-shot in some cases.")

print("\nFew-Shot:   Uses multiple diverse examples to establish clear patterns.")
print("            Best performance as it captures various writing styles in social media.")

print("=" * 100)
```

OUTPUT:

```
==========================================================================================
SOCIAL MEDIA POST CLASSIFICATION RESULTS
==========================================================================================

Zero-Shot Prompting Results:
------------------------------------------------------------------------------------------
1. ✓ Post: Just got the new product and it's amazing! 10/10 would recom...
   Predicted: Appreciation    | Actual: Appreciation

2. ✓ Post: Why is your customer service so terrible? Been waiting 2 hou...
   Predicted: Complaint       | Actual: Complaint

3. ✓ Post: Check out our summer sale - 50% off everything! Link in bio...
   Predicted: Promotion       | Actual: Promotion

4. ✓ Post: How do I track my order?...
   Predicted: Inquiry         | Actual: Inquiry

5. ✓ Post: This broke after one day. Total waste of money....
   Predicted: Complaint       | Actual: Complaint

6. ✓ Post: Love this brand so much! They always deliver quality...
   Predicted: Appreciation    | Actual: Appreciation

7. ✓ Post: Limited time offer! Buy now and get free shipping...
   Predicted: Promotion       | Actual: Promotion

8. ✓ Post: Can someone help me with my account settings?...
   Predicted: Inquiry         | Actual: Inquiry

9. ✗ Post: Worst experience ever. Never buying again!...
   Predicted: Promotion       | Actual: Complaint

10. ✓ Post: Thanks for the quick response! Problem solved...
   Predicted: Appreciation    | Actual: Appreciation

11. ✓ Post: New collection just dropped! Shop now before it sells out...
   Predicted: Promotion       | Actual: Promotion

12. ✓ Post: Does this product come in other colors?...
   Predicted: Inquiry         | Actual: Inquiry


One-Shot Prompting Results:
------------------------------------------------------------------------------------------
1. ✓ Post: Just got the new product and it's amazing! 10/10 would recom...
   Predicted: Appreciation    | Actual: Appreciation

2. ✓ Post: Why is your customer service so terrible? Been waiting 2 hou...
   Predicted: Complaint       | Actual: Complaint

3. ✓ Post: Check out our summer sale - 50% off everything! Link in bio...
   Predicted: Promotion       | Actual: Promotion

4. ✓ Post: How do I track my order?...
   Predicted: Inquiry         | Actual: Inquiry

5. ✓ Post: This broke after one day. Total waste of money....
   Predicted: Complaint       | Actual: Complaint

6. ✓ Post: Love this brand so much! They always deliver quality...
   Predicted: Appreciation    | Actual: Appreciation
```

```
7. ✓ Post: Limited time offer! Buy now and get free shipping...
   Predicted: Promotion       | Actual: Promotion

8. ✓ Post: Can someone help me with my account settings?...
   Predicted: Inquiry         | Actual: Inquiry

9. ✗ Post: Worst experience ever. Never buying again!...
   Predicted: Promotion       | Actual: Complaint

10. ✓ Post: Thanks for the quick response! Problem solved...
   Predicted: Appreciation    | Actual: Appreciation

11. ✓ Post: New collection just dropped! Shop now before it sells out...
   Predicted: Promotion       | Actual: Promotion

12. ✓ Post: Does this product come in other colors?...
   Predicted: Inquiry         | Actual: Inquiry


Few-Shot Prompting Results:
----------------------------------------------------------------------------------------------------
1. ✓ Post: Just got the new product and it's amazing! 10/10 would recom...
   Predicted: Appreciation    | Actual: Appreciation

2. ✓ Post: Why is your customer service so terrible? Been waiting 2 hou...
   Predicted: Complaint       | Actual: Complaint

3. ✓ Post: Check out our summer sale - 50% off everything! Link in bio...
   Predicted: Promotion       | Actual: Promotion

4. ✓ Post: How do I track my order?...
   Predicted: Inquiry         | Actual: Inquiry

5. ✓ Post: This broke after one day. Total waste of money....
   Predicted: Complaint       | Actual: Complaint

6. ✓ Post: Love this brand so much! They always deliver quality...
   Predicted: Appreciation    | Actual: Appreciation

7. ✓ Post: Limited time offer! Buy now and get free shipping...
   Predicted: Promotion       | Actual: Promotion

8. ✓ Post: Can someone help me with my account settings?...
   Predicted: Inquiry         | Actual: Inquiry

9. ✗ Post: Worst experience ever. Never buying again!...
   Predicted: Promotion       | Actual: Complaint

10. ✓ Post: Thanks for the quick response! Problem solved...
   Predicted: Appreciation    | Actual: Appreciation

11. ✓ Post: New collection just dropped! Shop now before it sells out...
   Predicted: Promotion       | Actual: Promotion

12. ✓ Post: Does this product come in other colors?...
   Predicted: Inquiry         | Actual: Inquiry
```

```
====================================================================================================
ACCURACY COMPARISON
====================================================================================================

Zero-Shot       Accuracy: 91.67% (11/12 correct)
One-Shot        Accuracy: 91.67% (11/12 correct)
Few-Shot        Accuracy: 91.67% (11/12 correct)

====================================================================================================

RESEARCH INSIGHTS:
====================================================================================================
Zero-Shot:  Classifies based solely on category definitions without examples.
            Good for quick classification but may struggle with informal language nuances.

One-Shot:   Uses a single example to guide classification.
            Provides minimal context but better than zero-shot in some cases.

Few-Shot:   Uses multiple diverse examples to establish clear patterns.
            Best performance as it captures various writing styles in social media.
====================================================================================================
```